

Institute of Actuaries of India

Subject SP5 –Investment and Finance

November 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**Risks involved in infrastructure investments.**

The risks of an investment in infrastructure may be generally divided into:

- those specific to the infrastructure asset and
- those affecting the broader asset class.

[0.5]

Asset specific risks

The asset specific risks encompass risks pertaining to the design, construction and operation of the infrastructure asset.

[0.5]

The asset class risks include:

- a) Market / Economic risk

[0.5]

This is important when consumers can choose alternative services such as with toll roads, railways and ports.

One element of market risk is demand risk – ie the overestimation of customer demand and “willingness to pay” for the proposed infrastructure.

- b) Regulatory risk and Political risk.

[0.5]

A lack of clarity in Government policies and selection procedures has created political risk – a key risk in infrastructure development.

- c) Operating risk

[0.5]

This may arise due to underestimation of operating costs. If the pricing of infrastructure services is regulated by the government, the burden of underestimation of operating costs cannot be passed on entirely to the users.

Infrastructure assets may also be exposed to large-scale disasters, such as the fire in the Channel Tunnel in 2008.

The asset specific risks will largely depend on the maturity of the asset. For example, in the construction phase, there is considerable risk associated with the construction process.

[1]

In the initial high-risk planning phase only equity capital is suitable for financing. The construction phase may be financed by a combination of equity and debt with guarantees.

[1]

Importantly, a key feature of infrastructure assets is that as an asset matures its risk profile declines and valuation increases, all other things remaining equal.

[1]

Broader risks

Of the more generic risks affecting the infrastructure asset class, perhaps the most pertinent is interest rate risk. The prevailing level of interest rates can have an impact on the discount rates applied to the valuation of infrastructure investments, and on the debt portion of the investment structure.

[1]

Over the medium to longer term however, any initial fall in value as interest rates rise is mitigated as revenue from the underlying asset grows. Generally, revenue increases are derived from inflation-linked pricing increases and the volume increases that occur in a growing economy.

[1]

Of specific relevance to infrastructure projects is foreign exchange risk. Given that infrastructure projects involve costs and revenues in the local currency, the foreign exchange exposure taken for such investments, especially in the nature of off-shore debt, can prove to be risky. [1]

[Max. 6 Marks]

Solution 2:

Principal, L = 1,00,00,00,000
 Fixed Rate, R = 6%
 Forward swap Rate, F = 7%
 Volatility = 0.25
 Strike date is at time = 2 [1]

Annuity factor = $\exp(-3 \times 6.50\%) + \exp(-4 \times 6.60\%) + \exp(-5 \times 6.75\%) + \exp(-6 \times 7.00\%) = 2.9614$ [1]

$d_1 = 0.6128$ [1.5]

$d_2 = 0.2592$ [1.5]

$\Phi(d_1) = 0.7300$ [1]

$\Phi(d_2) = 0.6023$ [1]

Value of swap option = $L \times \text{Annuity factor} \times [F \times \Phi(d_1) - R \times \Phi(d_2)] = 4.43 \text{ crore}$ [1]

[Max. 8 Marks]

Solution 3:

i) *Switch from equities to bonds*

Revisions / rebalancing the current portfolio will incur transaction costs (brokerage commission, price impacts, bid-offer spreads and, possibly, tax liabilities). There are also costs of management and administration. [0.5]

Transaction costs may be reduced by implementing the switch in stages, rather than attempting it immediately; this will reduce any price impacts. [1]

To get immediate exposure in bonds for the new fund, we can long bond forwards or buy call options on bonds. [1]

Transaction costs may be eliminated by investigating exchange of shares and bonds at market price between the new fund and other funds to the extent the target allocations of other funds allow; this may be considered as partial advancing of the quarterly rebalancing exercise. [1]

Transaction costs may be eliminated for equity sales from the new fund by using the investment of cash-flows from other unit-linked products / funds. [0.5]

New net premiums of other unit-linked products and investment income from other funds can be used to buy equities at market price from the new fund to the extent target fund equity allocation allows. [0.5]

Transaction costs may be eliminated for bond purchases into the new fund by using any redemption (like surrenders or maturities) from other funds. Bonds from those funds can be sold to the new fund at market price.

[0.5]

[Max.4]

ii) ***Liability hedging***

Liability hedging is where the assets are chosen in such a way as to perform in the same way as the liabilities

[0.5]

Absolute matching (of the timing and amounts of all the individual cash-flows) can be achieved to the extent 4.75 year zero coupon government bonds are available in required quantity in the bond market,

[0.5]

Coupon bearing government bonds can be used for immunization where assets are matched to liabilities by term in order to reduce interest rate sensitivity (to parallel movements in the yield curve).

[0.5]

Liability hedging is a halfway house between absolute matching and immunization where assets perform exactly like the liabilities in all states.

[0.5]

This requires bonds to be held that is equal in present value to the future payments discounted at bond yields (using the full yield curve). Therefore, only a partial hedge is possible if asset cover is less than 100%.

[1]

For the new fund, as a percentage of the single premiums, value of assets is approximately

$$76\% \left(= 95\% (50\% + 50\% (1 - 40\%)) \right)$$

[1]

and the present value of guaranteed liability is

$$75.82\% \left(= \frac{100\%}{(1.06)^{4.75}} \right).$$

[1]

Assets are just equal to value of liabilities.

Swaps permit hedging to be achieved without full asset cover being required, as they are a contract for difference rather than a funded asset.

[0.5]

However, using swaps has some difficulties associated with it like need to prepare ISDA documentation; counterparty risk; closing out etc.

[0.5]

Due to "gaps" between bond maturities, there may be a need to reinvest or disinvest bonds prior to maturity, and the hedge may therefore be imperfect.

[1]

If the tax status of the government bonds worsens, this will mean the assets are likely to be insufficient to meet the liability payments.

[0.5]

Due to the above factors, there may be some mark to market risks between the asset value of the bond portfolio and the present value of the liability payments discounted using the bond yield curve.

[1]

Insurer can invest in high yielding bonds.

If yield curve is upward sloping then buy long dated bonds along with buying OTC put options on the bond portfolio with option term of 4.75 years to protect the fund value.

[1]

If yield curve is downward sloping then buy short-dated bonds and do rolling-over. However, this higher yield will not necessarily lead to a gain as reinvestment terms are uncertain and the "roll over" process may lead to higher transaction costs in the long term.

[1]

[Max. 8]**[12 Marks]**

Solution 4:

i)

- The inflation would have been within control, so the RBI would have reduced rates to spur economy growth.
- Lower interest rates would translate to lower cost of borrowing and hence encourage spending by consumers.
- Also, any investment in banks would yield lower interest payment resulting in reducing the incentives to save.
- Reduction in interest rate would result in lower EMI on the loans and hence higher money to spend.
- Lower cost of borrowing would encourage investment in capital projects.
- Lower interest rates make it more attractive to buy assets such as housing.
- Lower interest rates tend to be unattractive for foreign investments thereby decreasing the Indian currency's relative value.
- This would make the export cheaper and import costlier thereby increasing the volume of export and reducing the volume of import.

[8]

ii)

- The transmission of interest rates might have been sub-optimal.
- In absence of any external interest rates benchmark, banks might not have passed the significant part of benefits of rate cut to its customers.
- Thereby, the intended objectives of rates cut might not have been achieved.
- With an external benchmark, any rate cut by Central bank could be optimally passed on to the Customers.

[4]

[12 Marks]

Solution 5:i) **Portfolio theory efficient frontier**

a) A portfolio is efficient if there is no other portfolio with either a higher mean and the same or lower variance, or a lower variance and the same or higher mean. The efficient frontier is the set of efficient portfolios in E-V (or E- σ) space.

[1]

b) Optimal portfolio is the portfolio that maximizes the investor's expected utility as a function of the mean and variance of investment returns.

[1]

ii) **Plot efficient frontier**

Minimum variance portfolio (MV): proportion of bonds X_B is given by

$$X_B = \frac{V_E - C_{BE}}{V_B - 2C_{BE} + V_E} = \frac{0.21^2 - 0.007875}{0.15^2 - 2(0.007875) + 0.21^2} = 0.7124 = 71.24\%$$

$$X_E = 1 - X_B = 1 - 0.7124 = 0.2876 = 28.76\%$$

[2]

$$E = X_B E_B + X_E E_E \text{ and } \sigma = \sqrt{X_B^2 \sigma_B^2 + X_E^2 \sigma_E^2 + 2X_B X_E C_{BE}}$$

$$E_{MV} = 0.7124(8.1) + 0.2876(12.6) = 9.39\%$$

[0.5]

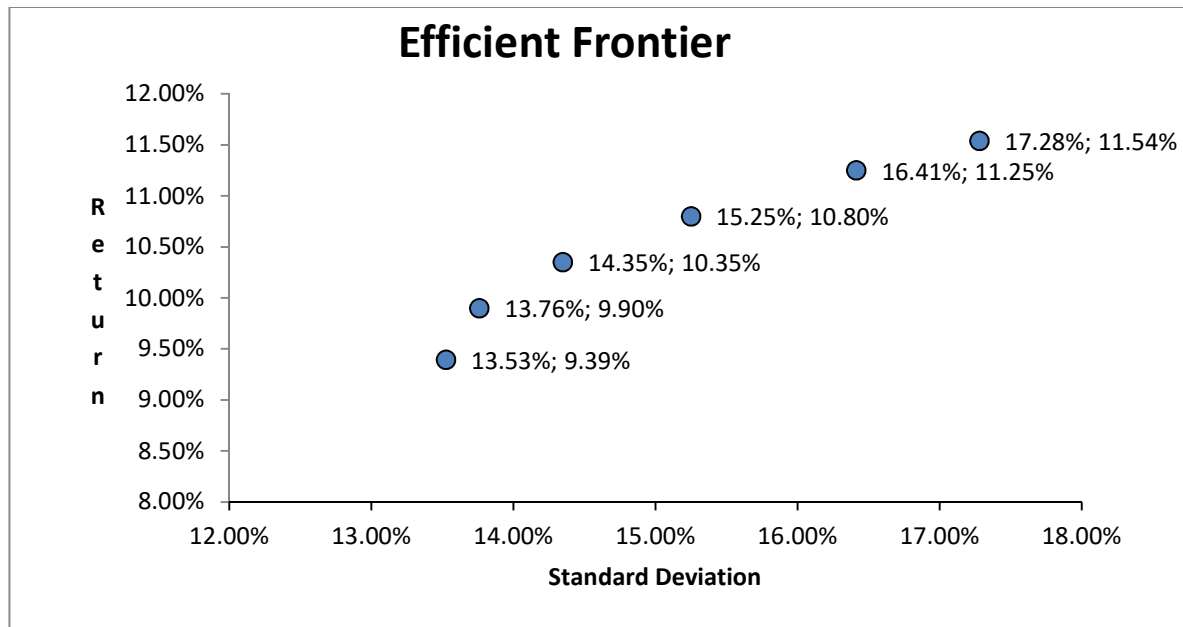
$$\sigma_{MV} = \sqrt{0.7124^2(0.15^2) + 0.2876^2(0.21^2) + 2(0.7124)(0.2876)(0.007875)} = 13.53\%$$

[1]

For Optimal portfolio (M), given $X_B = 23.54\%$; $X_E = 76.46\%$
 $E_M = 11.54\%$; $\sigma_M = 17.28\%$ from the table provided

We know that minimum variance portfolio (MV) and optimal portfolio (M) lie on efficient frontier. For two assets cases, if we vary X_B at MV towards M, we will get efficient portfolios with higher returns and higher standard deviation. [1]

In addition to MV and M portfolios, we may consider four portfolios having intermediate weights of X_B between that of MV and M i.e. $X_B = \{30\%; 40\%; 50\%; 60\%\}$ [0.5]



[2]

[7]

iii) **Market price of risk**

$$\text{Market price of risk} = \frac{E_M - R_{fr}}{\sigma_M} = \frac{11.54 - 6.00}{17.28} = 32.06\%$$

[1]

iv) **Risk adjusted returns**

$$\text{Sharpe measure} = \frac{E_P - R_{fr}}{\sigma_P}$$

[0.5]

$$\text{For Manager X} = \frac{16.00 - 6.00}{22.00} = 0.4545 \text{ for Manager Y} = \frac{22.00 - 6.00}{38.00} = 0.4211$$

[1]

Excess return with pre-specified standard deviation = $E_P - E_{BM}$; E_{BM} is benchmark return,
 $E_{BM} = R_{fr} + (\text{Market price of risk})(\sigma_P)$

[0.5]

For Manager X, excess return = $16.00 - 6.00 - 0.3206(22.00) = 0.0295 = 2.95\%$

For Manager Y, excess return = $22.00 - 6.00 - 0.3206(38.00) = 0.0382 = 3.82\%$

[1]

Comment

According to Sharpe measure, Manager X performance is better than that of Manager Y whereas comparing excess returns with pre-specified standard deviation shows Manager Y performance is better than that of Manager X. [1]

Sharpe measure is more appropriate than excess return with pre-specified standard deviation. A higher excess return with pre-specified standard deviation may be misleading where higher expected returns are coming by taking relatively higher risk. [1]

Consider an investor has invested all capital with the two fund managers X and Y in certain proportions giving some expected return. Suppose the investor intends to have a unit additional expected return by borrowing at risk free rate and invest that amount with any one manager. The new portfolio standard deviation would increase more if Manager Y is chosen than choosing Manager X implying Manager X has superior performance which is correctly captured by Sharpe measure. [2]

[6]

[16 Marks]

Solution 6:

i) **Strategy for bond Manager A**

Anomaly switching involves moving between stocks with similar volatility, thereby taking advantage of temporary anomalies in prices / yields. [0.5]

Policy switching is a more risky approach that involves taking a view on future changes in shape / slope or level of the yield curve and moving into bonds with quite different durations. [0.5]

To capitalize on expectations concerning a factor:

Factor	Manager A should make
changes in the level of yield curve	a policy switch, lengthening duration if rates are expected to fall, and shortening duration if rates are expected to rise
changes in the slope of yield curve	a policy switch, lengthening duration if slope of yield curve is expected to fall, and shortening duration if slope of yield curve is expected to rise
changes in yield spreads between sectors	an anomaly switch, buying bonds in the sector for which yields are expected to fall relative to other bonds and selling bonds in the sector for which yields are expected to rise relative to other bonds.
changes in yield of a particular instrument	an anomaly switch, buying the instrument if its yield is expected to fall relative to other similar bonds and selling the instrument if its yield is expected to rise relative to other similar bonds.

[1 mark: first two rows; 1.5 mark: last two rows]

[Max. 5 Marks]

ii) **Strategy for equity Manager B**

To hedge all of the exposure of equity portfolio to market movements, the appropriate number of derivatives with one month to expiry is determined by the ratio of the portfolio value to the value of the individual derivative used, multiplied by the optimal hedge ratio. [0.5]

Strategy		By next month value of equities

	Initial cost / capital required	goes up	goes down
Buy required number of put options	Small cost of premiums on put options	Full upside slightly reduced to the extent of premiums paid for put options that became worthless	Losses offset by gains on put options, loss capped to premiums paid for the put options
Buy required number of call options and short same number of futures	Substantial capital for the margin requirements for futures along with small cost of premium on call options	Full upside slightly reduced to the extent of premiums paid for call options and cost of maintaining futures position, loss on futures is offset by gains on call options	Losses offset by gains on futures, loss capped to premiums paid for the call options that became worthless and cost of maintaining futures

[2 marks/ row]

If all equity fund managers have the same view and rush to protect their portfolio value by buying put options, the premiums on put options would increase leading to significant disturbance in put-call parity. [0.5]

Given that the benefits of the two strategies are same, evaluate the costs of the two strategies by comparing cost of premiums paid on put options vs. the cost of premiums paid on call options plus cost of maintaining futures position. Choose the cheaper strategy. [1]

[Max. 5 Marks]

iii) **Performance Assessment**

Published bond / equity indices

The main advantage of assessing the performance relative to published indices is that it is relatively easy to do. By definition, the data for published indices is readily available - and it should be reliably accurate. [1]

A major potential disadvantage is the published index might not be appropriate. There may be no single index which is consistent with the objectives of the pension fund. [0.5]

It is often difficult to replicate the holdings in the market index without incurring substantial research / trading costs. Generally, this issue is more (less) severe with bond (equity) indices where constituents are in thousands (tens or hundreds). [1]

A market index may exhibit survivorship bias. Companies that have gone out of business or that have been delisted are removed from the index. This may result in a performance measure that overstates actual performance. [1]

A market index may exhibit double counting that arises because of a company and its subsidiary / associated company both being represented in the index. [0.5]

The chosen index may not represent the entire universe of securities. For example, the FTSE 100 Index represents 80% of U.K. equity market capitalization. [0.5]

If the chosen index is market capitalization weighted index, the index will have a large capitalization bias. This may be acceptable for equities but for bonds may not be sensible to invest more in a company's debt that has large outstanding debt. [0.5]

The chosen index may not be investable. There may be securities in the index of companies like tobacco processing companies that cannot be held in the portfolio due to the funds own restrictions on socially responsible investment. [0.5]

Competitor portfolios targeting 75th percentile

This type of comparison is appropriate if the funds being compared have the same objectives and the same factors influencing investment strategy. It also gives an indication of the cost or benefit of following a particular strategy, relative to that adopted by other funds. [1]

For example, it would be quite sensible to compare the returns achieved by different pension funds where the funds have the same liability profile (by type and term); similar solvency levels; similar size; similar mandates; similar levels of cash-flow. [1]

Considering the above parameters, there are very few other pension schemes that can be used for a valid comparison. In practice, the comparisons tend to be driven more by what data is available. It may be difficult to obtain all competitors data. [0.5]

However, it may be very inappropriate to compare the performance of funds that have very different investment objectives, [0.5]

It can be difficult to identify a universe of competitor portfolios appropriate for the investment style of the pension fund managers and selection of a universe for comparison involves some subjective judgment. [0.5]

The 75th percentile of competitor portfolios does not represent an “investable” portfolio; i.e, a portfolio manager is able to invest in target portfolio only after assessment period is over.

The benchmark is not constructed prior to the start of an assessment period; it is not specified in advance. [1]

This implies an ambiguous benchmark to fund managers. The names and weights of the securities constituting the benchmark are not mentioned clearly. [0.5]

Competitor portfolios may exhibit survivorship bias; competitors that have gone out of business are removed from the universe. This may result in a performance measure that overstates the actual performance. [0.5]

[Max. 8 Marks]

iv) *Alternative assessment method*

Benchmark portfolios can be constructed to reflect the objectives of the fund. They should also be constructed in such a way that the data necessary for comparisons is easily obtained.

They can therefore overcome the problems we identified with the other two forms of comparison. [1]

By having a benchmark portfolio that reflects the liabilities of the fund, the danger of giving the fund manager conflicting objectives is avoided. [0.5]

Assessing against a benchmark portfolio helps to identify some relative strengths and/or weaknesses of individual fund managers (e.g. in sector or stock selection). [1]

This is the most difficult performance measurement method to develop and calculate. [1]

For an evolving liability profile, the benchmark portfolio needs to be frequently updated, requiring substantial resources to perform ALM exercise [0.5]

There is a conflict of interest when the managers are involved in developing and calculating their benchmark portfolio. They may produce an easy to beat benchmark, making their performance appear better than it actually is.

[1]

[Max. 4]

[22 Marks]

Solution 7:

i)

- Private debt is a debt capital market transaction that generally has covenant features similar to a bank loan and is often used as an alternative to bank funding.
- Unlike a traditional public debt market transaction which is distributed to a wide range of investors and can trade in an active secondary market, private debt is not actively traded. Typically, it will be marketed to a much smaller number of long-term “buy and hold” investors.
- When weighing up investment opportunities, private debt investors tend to be very focused on covenant protection since illiquid markets mean they may be unable to sell their private debt investments in the event of a credit downturn.
- Although a private debt issuer almost inevitably cedes covenants to investors, this is more than compensated for by the fact that it is able to issue capital market debt without acquiring a formal long-term debt rating.
- From a treasurer’s perspective, the private debt market represents a major capital source with relatively competitive pricing.
- Indeed, the vast majority of private debt transactions are undertaken by treasurers who want to refinance existing term loans from banks, but who would rather not go to the trouble and expense of obtaining a credit rating.
- However, a treasurer may be able to improve his covenant bargaining position (or improve his pricing) by obtaining a private rating from an external rating agency.
- Private debt funding is generally medium-term to long-term (in excess of three years). Although private debt can be repaid early, this will require the issuer to make a costly “make whole” premium.

[1 mark each]

[Max. 6]

ii) **Private Debt**

The term private debt refers to loan capital issued by companies that is not publicly listed and traded on a stock exchange.

[1]

In case of private debts, XYZ would have to agree for some covenants. The covenants might restrict his intended expansion style to some extent so due considerations should be given to the nature of covenants before accepting them.

[1]

Further, the cost of private debt would be higher than the bank loans because investors would like to be compensated higher and commensurate with the greater risks involved – due to a lack of information and marketability.

[1]

The management rights in case of private debts remain with Mr. XYZ.

[1]

Private Equity

In case of private equity, he would have to share management rights and there wouldn't be any additional interest cost to be serviced by Mr. XYZ. [1]

Two forms of private equities might be more suitable in this case:

- Developmental capital – growth or expansion working capital for mature businesses in need of product extension and/or market expansion.

This development capital may be provided en-route to a public issue when the company is sufficiently large and profitable.

It can be used to fund organic growth, ie the expansion of existing lines of production, or new projects in different markets. [1.5]

- Venture capital – capital for businesses in the conceptual stage or where products are not developed and revenues and/or profits may not have been achieved.

As the Company has been successful in attracting customers and been showing promising growth, venture capitalists might be interested in investment.

Venture capitalists are rarely just “silent partners”, investing their capital and then having no involvement in the operation of the business.

Rather, their expertise, their guidance, and possibly their contacts, are often crucial to the success of the business. [1.5]

[8]

iii)

- Tax holiday can be provided to start-ups.
This incentive would help businesses meet their capital requirements while setting up.
- Exemption from tax on long-term capital gains.
This would encourage investors to invest in start-up firms.
- Tax exemption on investments above the fair market value

It is often difficult to determine the fair market value of shares of start-ups. In majority of the cases, the capital investment is higher than the fair market value thereby resulting into being taxed in absence of tax exemption.

[1 mark for each incentive and 1 for each reason]

[4]

iv)

- Anchoring: [1]

The success of a start-up is dependent upon many factors including the profile and experience of management. Accordingly, the probability of success for each start-up would be different. [1]

But in this case, the low probability of success of start-up published in the report has influenced Mr. XYZ that his start-up might also fail. [1]

- Prospect theory: [1]

Prospect theory is a theory of how people make decisions when faced with risk and uncertainty. [1]

In this case, the expected income from start-up might have been greater than the interest income from FD but as significant risk is involved in the start-up so he is getting confused when presented with an option of guaranteed income from FD. [1]

[6]

[24 Marks]
